Genericization and Nominalization: Text Mining Scholarly Citational Practices

Warner, Matt
mattgw@stanford.edu
Stanford University, United States of America

Nomura, Nichole
nnomura@stanford.edu
Stanford University, United States of America

Keane, Gabi
gab03@stanford.edu
Stanford University, United States of America

Thong, Carmen
tcarmen@stanford.edu
Stanford University, United States of America

Algee-Hewitt, Mark
malgeehe@stanford.edu
Stanford University, United States of America

We track two inter-related phenomena, genericization and nominalization, by which scholars' names become detached from their concepts and ideas as they circulate through scholarship. By *genericization* we refer to the processes by which ideas such as "intersectionality" and "orientalism", which are at first strongly associated with individual scholars (respectively, in this case, Kimberlé Crenshaw and Edward Said) come to be used "generically", without reference to the scholars whose work originated these concepts. Conversely, *nominalization* (closely related to the practice scientometric literature terms *eponymous adjectivization*) refers to the process by which a scholar's name comes to serve (often adjectivally) as a shorthand for their ideas, for example, the *Foucauldian* concept of discourse.

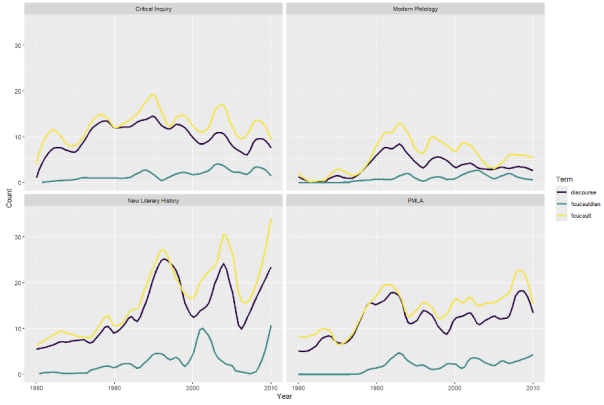

Figure 1: absolute frequencies of the terms "discourse", "foucauldian" and "foucault" in four journals over the period 1980-2010.

Figure 1, for example, shows the relative frequencies of the (case-insensitive) terms "discourse", "foucault" and "foucauldian" over the period 1980-2010, showing the strong correlation of "discourse" with Foucault's work, and the different trajectories this concept has followed in different domains of literary studies. In all four journals shown here, however, we see the lag between the popularization of "discourse" (and presumably, other terms linked to Foucault's scholarship) and the emergence of "Foucauldian" as a term. While these trajectories will of course vary according to the scholar in question, we propose that fundamentally, although genericization and nominalization are conceptually opposite phenomena, they have developed in ways strongly distorted by structural forces such as gender, with, for example, women and gender-minority scholars much less likely to be nominalized, and, we believe, their work correspondingly more likely to be genericized.

Naming practices have been most thoroughly studied in the sciences and scientometrics, where there is longstanding interest in the affixation of scientists' names to laws, theories, constants, lemmas, etc. Since the same phenomenon may be discovered by multiple individuals, and since eponyms provide memorable, but non-informative, names, this practice is not without controversy, particularly in the field of medicine (Whitworth 2007; Woywodt & Matteson 2007). Scholarship in this area has examined the role that eponyms play in the social systems of science, as both reward and memorial (Kay 1973; Garfield 1983) as well as the controversies surrounding their use (Stigler 1980; Wallace 2000; Woywodt & Matteson 2007). Of particular relevance for this project, Száva-Kováts 1994 considers the use of eponyms *without* accompanying citations,

while McCain 2012 considers "obliteration by incorporation", a process similar to our "genericization," and, from a text-mining perspective, Cabanac 2014 treats the semi-supervised extraction of eponyms from article text via regular expressions.

The case of scientific eponyms differs from the use of scholars' names in the humanities in a number of ways, however. In english-langauge scholarship, humanities eponyms are often constructed as adjectives using suffixes such as "-ian", while in scientific and medical fields noun phrases using the possessive are more common (e.g., "Euler's Number"). Beyond this grammatical difference, eponyms are less common in humanities fields and, not unlike the occasional use of adjectival forms in the sciences ("Newtonian physics"), generally refer to a methodology or mode of inquiry, rather than a particular fact or principle. This means that the eponym stands for a concept often articulated across a large body of work, and in some cases it can be difficult to point to any one authoritative instantiation of the idea in the eponymous scholar's work.

We propose to explore genericization and nominalization in english-language literary studies scholarship using two distinct corpora: first, a collection of journal articles (obtained via scholarly database request) drawn from both a) well-known widely-scoped English-language journals such as *PMLA, Modern Language Quarterly, Critical Inquiry,* etc., as well as from b) notable journals representing a wide variety of subfields of literary studies (e.g., *Victorian Studies, Renaissance Quarterly, Digital Humanities Quarterly),* and second, a collection of several thousand monographs drawn from a mass digitization project (this corpus was detailed in our 2024 ADHO presentation "Slow, Painful and Expensive: Current Challenges in Corpus Construction"). With both corpora, our selections will be exploratory, and should not be seen as representative: for example, we hope that the field-specific journals will allow us to contrast a more general view against a variety of scholarly domains whose practices we expect to vary. While our monographs are drawn from a multilingual collection, our journal articles are entirely english-language, and this stage of the project will focus on scholarship written in English (albeit on literature written in a variety of languages).

Technically, our project seeks to understand the process by which genericization and nominalization occur by measuring the evolving distance (number of words) between the names of scholars and the key terms they introduce. In the case of genericization, we hypothesize that terms (e.g. "intersectional") will originally be tightly linked to their respective authors through citational practices. Over time, this distance will increase as the governing author moves from citation to reference and then absence. In the case of nominalization, we hypothesize a similar trajectory away from the author's name, with the adjectival form supplanting many of the uses of the author's names (i.e., a reference or a formal citation). To measure this phenomenon, we track the relative frequencies of the scholar's name and the nominalization over time. Initially, we hypothesize that any use of a nominalized form will be accompanied by the use of the author's name in a citation or reference (if not in the same sentence than at least elsewhere in the same text), while over time it will become more common for the nominalized form to be used alone.

Genericization and nominalization—two ways, respectively, for a scholar's name to enter or exit the scholarly discourse—are a vital part of the scholarly systems by which we recognize contributions to the scholarly field, and comparing these processes as they differ not merely from journal to journal but from scholar to scholar will allow us to build a better understanding of the different and gendered ways that different scholars' names are used: whose name is becomes too important to forget? which scholars are so well known that they need not be formally cited?

Bibliography

Cabanac, Guillaume. (2014). Extracting and quantifying eponyms in full-text articles. *Scientometrics*, *98*(3), 1631–1645.

Kay, H. E. M. (1973). In praise of eponyms. *The Lancet, 302*(7840), 1256.

Garfield, Eugene. (1973). Uncitedness III—the importance of *Not* being cited. *Current Contents, 8*, 5–6.

Garfield, Eugene. (1983). What's in a name? the eponymic route to immortality. Current Contents: Social and Behavioral Sciences 14(47): 384-395.

McCain, Katherine. W. (2012). Assessing obliteration by incorporation: issues and caveats. *Journal of the American Society for Information Science and Technology, 63*(11), 2129–2139.

Stigler, Stephen M. (1980). Stigler's law of eponymy. In T. F. Gieryn (Eds.) *Transactions of the New York Academy of Sciences, vol 39* (pp 147–157).

Száva-Kováts, Endre. (1994). Non-Indexed Eponymal Citedness (NIEC): First fact-finding examination of a phenomenon of scientific literature. *Journal of Information Science, 20*(1), 55–70.

Wallace, Daniel. J., & Weisman, Michael. (2000). Should a war criminal be rewarded with epony-

mous distinction? The double life of Hans Reiter (1881–1969). *Journal of Clinical Rheumatology, 6*(1), 49–54.

Whitworth, Judith. A. (2007). Should eponyms be abandoned? No. *British Medical Journal, 335*(7617), 425.

Woywodt, Alexander., & Matteson, Eric. (2007). Should eponyms be abandoned? Yes. *British Medical Journal, 335*(7617), 424.